INCH-POUND

MIL-DTL-32124B(OS)
3 June 2026
SUPERSEDING
MIL-DTL-32124A(OS)
13 October 2006

DETAIL SPECIFICATION

PROPELLANT, NSWC/IH-BC-15, NSWC/IH-BC-16, NSWC/IH-BC-17, NSWC/IH-BC-20, NSWC/IH-BC-21, AND NSWC/IH-BC-22

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers six types of propellant, designated as follows:

1.2 Types.

a. Type I – NSWC/IH-BC-15
b. Type II – NSWC/IH-BC-16
c. Type III – NSWC/IH-BC-17
d. Type IV – NSWC/IH-BC-20
e. Type V – NSWC/IH-BC-21
f. Type VI – NSWC/IH-BC-22

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-82655 - Ferric Oxide, Technical
MIL-L-82661 - Lecithin, Technical
MIL-A-82667/2 - Ammonium Perchlorate, Conditioned
MIL-S-82858 - Silicon Dioxide, Microfine

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 1377

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MIL-C-85498 - Curing Agents, Dimeryl-Di-Isocyanate and Isophorone Di-Isocyanate

NOTE: MIL-C-85498 is inactive for new design and will be eventually replaced.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) SPECIFICATIONS (CAGE CODES 53711 AND 10001)

WS 19733 - Imine Curing/Bonding Agents

WS 19738 - Antioxidant, T-Butylphenol type

WS 19740 - Dioctyl Adipate

WS 20700 - Hydroxyl Terminated Polybutadiene

OS 10170 - Aluminum Powder

(Copies of these publications are available by email request to CommandStandards@navy.mil.)

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS (CAGE CODE 30003)

673AS131 - Liner Filler

(Copies of this publication are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

CHEMICAL PROPULSION INFORMATION AGENCY (CPIA)

Publication No. 21 - JANNAF Solid Propellant Physical Behavior Manual

(Application for copies should be addressed to Chemical Propulsion Information Agency, The Johns Hopkins University Applied Physics Laboratory, Johns Hopkins Road, Laurel, MD 20707.)

ASTM INTERNATIONAL

ASTM D792 - Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement

ASTM D2240 - Standard Test Method for Rubber Property – Durometer Hardness

(Copies of these publications are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a first article sample (see 6.3) of the cured propellant consisting of one batch (see 6.7.1) shall be manufactured using the methods and procedures proposed for production (see 4.3.1).

3.1.1 Aging characteristics. Unless otherwise specified (see 6.2), the propellant shall be tested in accordance with the applicable end item specification to ensure compliance to table II.

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3.2 Material. The material shall be a solid composite propellant as described herein.

3.3 Chemical composition and ingredients. The chemical composition and ingredients shall be in accordance with table I and sections 3.3.2, 3.3.3, and 3.3.4. As specified (see 6.2), a record of the ingredient weights and ingredient lot/batch number for each batch of propellant shall be submitted.

TABLE I. Chemical composition of NSWC/IH-BC-15, NSWC-IH-BC-16, NSWC-IH-BC-17, and NSWC-IH-BC-20.

Ingredients	Specification/Type	NSWC-IH-BC-15 Percent by Weight	NSWC-IH-BC-16 Percent by Weight	NSWC-IH-BC-17 Percent by Weight	NSWC-IH-BC-20 Percent by Weight
HTPB	WS 20700, type 1	15.97±0.1597	14.46±0.1446	13.06±0.1306	10.4±0.104
Isophorone Di-Isocyanate	MIL-C-85498, type 2	1.10±0.0110	1.11±0.0111	1.01±0.0101	0.8±0.0080
Antioxidant	WS 19738	0.17±0.0017	0.17±0.0017	0.17±0.0017	0.15±0.0015
Dioctyl Adipate	WS 19740	-	1.50±0.0150	3.00±0.03	3.00±0.03
HX-752	WS 19733, type 1	0.30±0.003	0.30±0.003	0.30±0.003	0.30±0.003
Cab-O-Sil, Microfine SiO_2	MIL-S-82858	0.50±0.005	0.50±0.005	0.50±0.005	-
Liner Filler	673AS131	0.10±0.001	0.10±0.001	0.10±0.001	-
Ferric Oxide	MIL-DTL-82655, class 2	1.50±0.015	1.50±0.015	1.50±0.015	2.00±0.02
Aluminum Powder	OS 10170B, class 1	2.00±0.02	2.00±0.02	2.00±0.02	2.00±0.02
Ammonium Perchlorate	MIL-A-82667/2, type III, class 1	78.36±0.7836 (2 sizes): 200µ [1/], 55µ	78.36±0.7836 (2 sizes): 200µ [1/], 55µ	78.36±0.7836 (2 sizes): 200µ [1/], 55µ	81.35±0.8135 (2 sizes): 200µ [1/], 30µ

NOTE:

[1/] Particle size shall be ground from the same 200µ stock, and as listed, is the 50 percent nominal value by weight.

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TABLE I. Chemical composition of NSWC/IH-BC-21 and NSWC/IH-BC-22 – Continued.

Ingredients	Specification/Type	NSWC-IH-BC-21 Percent by Weight	NSWC-IH-BC-22 Percent by Weight
HTPB	WS 20700, type 1	11.65±0.1165	10.87±0.1087
Isophorone Di-Isocyanate	MIL-C-85498, type 2	0.90±0.009	0.84±0.0084
Antioxidant	WS 19738	0.15±0.0015	0.15±0.0015
Dioctyl Adipate	WS 19740	3.00±0.03	2.70±0.027
HX-752	WS 19733, type 1	0.30±0.003	0.30±0.003
Cab-O-Sil, Microfine SiO_2	MIL-S-82858	-	-
Liner Filler	673AS131	-	-
Lecithin	MIL-L-82661	-	0.30±0.003
Ferric Oxide	MIL-DTL-82655, class 2	2.00±0.02	2.50±0.025
Aluminum Powder	OS 10170B, class 1	2.00±0.02	5.00±0.05
Ammonium Perchlorate	MIL-A-82667/2, type III, class 1	80.00±0.8 (2 sizes): 200µ 1/, 10µ	77.34±0.7734 (3 sizes): 200µ 1/, 13µ, 6µ

NOTE:

1/ Particle size shall be ground from the same 200µ stock, and as listed, is the 50 percent nominal volume by weight.

3.3.1 Weight tolerances. Each ingredient shall be weighed to ensure compliance with table I prior to mixing. Weights shall be measured with equipment having a minimum accuracy of ±1.0 percent of the nominal ingredient weight specified in table I.

3.3.2 Ammonium perchlorate (AP) ratio of particle sizes. Control of the propellant burn rate shall be maintained by varying the ratio of weight fractions of the unground (coarse) and ground (fine) AP (see table I). The weight fractions are the weight of the coarse AP or fine AP divided by the total weight of the coarse and fine AP. The ratio of the coarse (200µ) AP/fine (55µ) AP for NSWC/IH-BC-15, NSWC/IH-BC-16 and NSWC/IH-BC-17 shall be nominally 30/70 but may be adjusted as needed. The ratio of the coarse (200µ) AP/fine (30µ) AP for NSWC/IH-BC-20 shall be nominally 38/62 but may be adjusted as needed. The ratio of the coarse (200µ) AP/fine (10µ) AP for NSWC/IH-BC-21 shall be nominally 40/60 but may be adjusted as needed. The ratio of the coarse (200µ) AP/fine (13µ) AP/fine (6µ) AP for NSWC/IH-BC-22 shall be nominally 16/41/43 but may be adjusted as needed.

3.3.3 Binder.

3.3.3.1 Binder, NSWC/IH-BC-15. Control of the physical properties of the propellant shall be maintained by varying the amounts of Hydroxyl-Terminated Polybutadiene (HTPB) polymer (WS 20700), Isophorone Di-Isocyanate (IPDI) (MIL-C-85498) and HX-752 (WS 19733) bonding agent. The amounts of HTPB polymer and IPDI shall be determined by adjusting the ratio of chemical equivalents of the curative (IPDI) to the chemical equivalents of the HTPB polymer. The value of this ratio shall be nominally 0.80/1.0 based on the functional equivalence values from the vendor's ingredient certifications. The amount of the HX-752 shall be nominally 0.30 percent and can be adjusted as needed to improve cold temperature strain properties. Total amounts of the HTPB polymer, IDPI, and HX-752 shall remain constant at 17.37±1 percent.

3.3.3.2 Binder, NSWC/IH-BC-16. Control of the physical properties of the propellant shall be maintained by varying the amounts of HTPB polymer conforming to WS 20700, IPDI (MIL-C-85498) and HX-752 bonding agent. The amounts of HTPB polymer and IPDI shall be determined by adjusting the ratio of chemical equivalents of the curative (IPDI) to the chemical equivalents of the HTPB polymer. The value of this ratio shall be nominally

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0.90/1.0 based on the functional equivalence values from the vendor's ingredient certifications. The amount of the HX-752 shall be nominally 0.30 percent and can be adjusted as needed to improve cold temperature strain properties. Total amounts of the HTPB polymer, IDPI, and HX-752 shall remain constant at 15.87±1 percent.

3.3.3.3 Binder, NSWC/IH-BC-17. Control of the physical properties of the propellant shall be maintained by varying the amounts of HTPB polymer conforming to WS 20700, IPDI (MIL-C-85498) and HX-752 bonding agent. The amounts of HTPB polymer and IPDI shall be determined by adjusting the ratio of chemical equivalents of the curative (IPDI) to the chemical equivalents of the HTPB polymer. The value of this ratio shall be nominally 0.90/1.0 based on the functional equivalence values from the vendor's ingredient certifications. The amount of the HX-752 shall be nominally 0.30percent and can be adjusted as needed to improve cold temperature strain properties. Total amounts of the HTPB polymer, IDPI, and HX-752 shall remain constant at 14.37±1 percent.

3.3.3.4 Binder, NSWC/IH-BC-20. Control of the physical properties of the propellant shall be maintained by varying the amounts of HTPB polymer conforming to WS 20700, IPDI (MIL-C-85498) and HX-752 bonding agent. The amounts of HTPB polymer and IPDI shall be determined by adjusting the ratio of chemical equivalents of the curative (IPDI) to the chemical equivalents of the HTPB polymer. The value of this ratio shall be nominally 0.90/1.0 based on the functional equivalence values from the vendor's ingredient certifications. The amount of the HX-752 shall be nominally 0.30 percent and can be adjusted as needed to improve cold temperature strain properties. Total amounts of the HTPB polymer, IDPI, and HX-752 shall remain constant at 11.50±1 percent.

3.3.3.5 Binder, NSWC/IH-BC-21. Control of the physical properties of the propellant shall be maintained by varying the amounts of HTPB polymer conforming to WS 20700, IPDI (MIL-C-85498) and HX-752 bonding agent. The amounts of HTPB polymer and IPDI shall be determined by adjusting the ratio of chemical equivalents of the curative (IPDI) to the chemical equivalents of the HTPB polymer. The value of this ratio shall be nominally 0.90/1.0 based on the functional equivalence values from the vendor's ingredient certifications. The amount of the HX-752 shall be nominally 0.30 percent and can be adjusted as needed to improve cold temperature strain properties. Total amounts of the HTPB polymer, IDPI, and HX-752 shall remain constant at 12.85±1 percent.

3.3.3.6 Binder, NSWC/IH-BC-22. Control of the physical properties of the propellant shall be maintained by varying the amounts of HTPB polymer conforming to WS 20700, IPDI (MIL-C-85498) and HX-752 bonding agent. The amounts of HTPB polymer and IPDI shall be determined by adjusting the ratio of chemical equivalents of the curative (IPDI) to the chemical equivalents of the HTPB polymer. The value of this ratio shall be nominally 0.90/1.0 based on the functional equivalence values from the vendor's ingredient certifications. The amount of the HX-752 shall be nominally 0.30 percent and can be adjusted as needed to improve cold temperature strain properties. Total amounts of the HTPB polymer, IDPI, and HX-752 shall remain constant at 12.01±1 percent.

3.3.4 Formulation adjustment. The nominal values and nominal limits of the ingredient weight percentages (see table I), the curative to HTPB polymer ratio (see 3.3.3) were determined experimentally (see 6.4). Decisions to vary these ratios and ingredient weight percentages from their nominal values and nominal limits shall be based on ballistic and physical properties obtained from previous batch history, trends indicated by statistical control charts, or lot changes of raw materials. The nominal limits of these ratios and weight percentages shall not be criteria for accepting or rejecting a batch of propellant. Only the physical properties (see table II) and ballistic performance of the end item (see 6.1) shall be propellant acceptance/rejection criteria. Failure to meet the requirements of 3.4 and the ballistic performance of the end item (see 6.1) shall result in rejection of the propellant lot.

3.4 Physical properties. The physical properties of the cured propellant shall conform to table II when cured in accordance to 4.4.3. Test specimens shall be tested within 30 days after the end of cure. As specified (see 6.2), data from the tests specified in 4.5.4 through 4.5.8 shall be reported.

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TABLE II. Physical properties of the propellant.

Physical Property	Temp (±5 °F)	NSWC/IH-BC-15		NSWC/IH-BC-16		NSWC/IH-BC-17		NSWC/IH-BC-20	
		Min	Max	Min	Max	Min	Max	Min	Max
Maximum tensile stress (psi)	165	86.2	-	80	-	81.2	-	49	-
Strain at maximum stress (%)	-65	7.1	-	6	-	15.5	-	8	-
Heat of explosion (cal) (reference)	77	1125	1135	1125	1135	1125	1135	1280	1300
Shore hardness "A" (reference)	77	55	67	55	67	55	67	65	75
Density (lb/in^3)	77	0.059	0.061	0.059	0.061	0.059	0.061	0.0605	0.0625

TABLE II. Physical properties of the propellant – Continued.

Physical Property	Temp (± 5°F)	NSWC/IH-BC-21		NSWC/IH-BC-22	
		Min	Max	Min	Max
Maximum tensile stress (psi)	165	65	-	70	-
Strain at maximum stress (%)	-65	14	-	10	-
Heat of explosion (cal) (reference)	77	1125	1135	1280	1300
Shore hardness "A" (reference)	77	55	67	65	75
Density (lb/in^3)	77	0.0595	0.0615	0.0605	0.0625

3.5 Burn rates of the cured propellant. A burn rate test shall be conducted on the propellant cured in accordance with 4.4.3. As specified (see 6.2), data from the tests specified in 4.5.8 shall be reported. The burn rate shall be such as to meet the performance required of its intended use. Advisory burn rates are shown in tables III, IV, V, and VI (see 6.5).

3.6 Workmanship. The material shall be uniform in consistency and free from contamination, foreign matter, or any other defect that would prevent its use for the purpose intended.

TABLE III. Advisory burn rates of the propellant, NSWC/IH-BC-15, NSWC/IH-BC-16, and NSWC/IH-BC-17.

Pressure Versus Temperature	2000 psia (in/sec)		4000 psia (in/sec)		6000 psia (in/sec)		8000 psia (in/sec)	
	Min	Max	Min	Max	Min	Max	Min	Max
-65±5 °F	0.78	-	1.20	-	1.53	-	1.85	-
165±5 °F	-	1.04	-	1.57	-	2.01	-	2.51

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TABLE IV. Advisory burn rates of the propellant, NSWC/IH-BC-20.

Pressure Versus Temperature	1000 psia (in/sec)	2500 psia (in/sec)	3000 psia (in/sec)	4000 psia (in/sec)
	nominal	nominal	nominal	nominal
-65±5 °F	0.595	0.91	0.987	1.13
165±5 °F	0.759	1.14	1.25	1.43

TABLE V. Advisory burn rates of the propellant, NSWC/IH-BC-21.

Pressure Versus Temperature	1000 psia (in/sec)	2000 psia (in/sec)	3000 psia (in/sec)	4000 psia (in/sec)
	nominal	nominal	nominal	nominal
-65±5 °F	0.771	0.974	1.19	1.39
165±5 °F	0.939	1.154	1.443	1.707

TABLE VI. Advisory burn rates of the propellant, NSWC/IH-BC-22.

Pressure Versus Temperature	500 psia (in/sec)	1000 psia (in/sec)	1500 psia (in/sec)	2000 psia (in/sec)
	nominal	nominal	nominal	nominal
77±5 °F	0.901	1.35	1.481	1.645

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.5.

4.3 First article inspection. First article inspection shall be performed on cured propellant when a first article sample is required (see 3.1). This inspection shall include the tests of 4.3.1 through 4.4.3 and the examination of 4.5. Failure to pass any examination shall result in rejection of the first article sample.

4.3.1 First article sample. Unless otherwise specified (see 6.2), a sample (see 4.4.1) consisting of one batch of the propellant shall be manufactured in accordance to 3.1. Test samples in accordance with 4.4.3 shall be prepared from propellant drawn from the first article batch and forwarded to a designated activity (see 6.2) for first article inspection. As specified (see 6.2), a copy of the propellant mixing procedure and applicable casting procedure shall be furnished prior to delivery of the first article sample. The sample shall be tested to determine, prior to starting production, that the production methods are capable of producing propellant that complies with the technical requirements of the contract. No process changes shall be made, subsequent to the approval of the first article sample, without prior approval of the procuring activity. The remainder of the batch may be used to cast units for use in the first article sample assemblies required by the applicable end item specification.

Rejection of the propellant first article sample will result in rejection of any units cast from the first article batch. Failure to meet the requirements of 3.1.1 and 3.4 shall result in the rejection of the first article sample. Further production of the propellant, prior to approval of the first article sample, is at the contractor's risk.

4.4 Conformance inspection. Conformance inspection shall include the examination of 4.5 and the tests of 4.4.2 and 4.4.3. Failure to pass any examination or test shall result in rejection of the lot. As specified (see 6.2), test reports showing quantitative results of all conformance tests required by this specification for each lot (see 6.7.2) of propellant shall be furnished.

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4.4.1 Sample size. The conformance lot sample size from the propellant lot shall be of sufficient quantity to perform all examinations and tests necessary to verify the properties listed in table II and the applicable burn rates of tables III, IV, V, or VI.

4.4.2 Sampling for conformance inspection. A conformance inspection sample, consisting of sufficient material to allow performance of the conformance tests as specified in 4.5, shall be withdrawn from each batch of propellant. Samples shall be cast and cured in accordance to 4.4.3.

4.4.3 Sample preparation. Samples for first article and conformance inspections shall be cast, in the same manner used for production units, in pans or blocks of sufficient size and quantity to complete all tests in 4.5.

4.4.3.1 Curing. Propellant grains and samples shall be cured together in the same oven at 140±5 °F until two consecutive Shore "A" hardness measurements, taken at least 24 hours apart, show no more than a 2 durometer point increase (see 6.9).

4.5 Examination and tests. Unless an alternative inspection method is proposed, tests shall be performed using the apparatus and procedures specified herein. Alternate inspection methods shall be approved by the contracting activity prior to use in acceptance testing.

4.5.1 Visual examination. All samples shall be visually examined to verify conformance with the workmanship requirements of 3.6. Packaging shall be examined for conformance to section 5.

4.5.2 Aging characteristics. Unless otherwise specified (see 6.2), the propellant shall be tested in accordance with the applicable end item specification to ensure compliance to 3.1.1.

4.5.3 Chemical composition. As specified (see 6.2), a record of the ingredient weights and ingredient lot/batch number for each batch of propellant to the contracting activity shall be submitted. Failure to meet the requirements of 3.3, 3.4, and the ballistic performance of the end item (see 6.1) shall result in the rejection of the propellant lot.

4.5.4 Maximum stress and strain at maximum stress. The maximum stress and strain at maximum stress of the cured propellant shall be in accordance with CPIA Publication No. 21, class C specimen. The average properties of five specimens tested at each temperature shall meet the requirements specified in table II. Crosshead rate shall be 2.0 inches/minute. Test sample temperature conditioning time shall be 3 hours minimum.

4.5.5 Density. The test specimen shall be tested in accordance with method 510.3.1 of MIL-STD-286 and the test temperature shall be 77±5 °F and shall meet the requirements of 3.4. An alternative test that may be used is ASTM D792, method A-1, except isopropyl alcohol shall be used in place of water.

4.5.6 Hardness. Hardness shall be determined in accordance with ASTM D2240, type A, at a temperature of 77±5 °F. The average of five different hardness readings taken on a single sample of propellant shall be reported as the hardness.

4.5.7 Heat of explosion. The heat of explosion of the cured propellant shall be determined in accordance with method 802.1 of MIL-STD-286. The results of five determinations and their average shall be recorded for each lot.

4.5.8 Burn rate. The burn rate of the cured propellant shall be determined for each batch of propellant in accordance to method 803.1.1 of MIL-STD-286 for each pressure-temperature combination specified in tables III, IV, V, and VI.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Intended use. NSWC/IH-BC-16 propellant is intended for use in the CCU-22B/A and CCU-149/A Impulse Cartridges. NSWC/IH-BC-20 propellant is intended for use in the CKU-5C/A Aircraft Ejection Seat Catapult. NSWC/IH-BC-21 propellant is intended for use in the CKU-7A/A, CKU-10A/A, and CKU-11A/A Aircraft Ejection Seat Catapults. NSWC/IH-BC-22 propellant is intended for use in the CKU-10A/A and CKU-11A/A Aircraft Ejection Seat Catapults. These propellants were developed for use in the cartridges and the catapult, are used in military aircraft only, and have no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1 and 6.3).

c. Whether aging characteristics are required (see 3.1.1).

d. Chemical composition and ingredients (see 3.3 and 4.5.3).

e. Inspection activity.

f. Inspection conditions, if other than as specified (see 4.2).

g. Assigned activity for first article inspection (see 4.3.1).

h. Aging characteristics (see 4.5.2).

i. Packaging requirements (see 5.1 and 6.8).

j. Lot size, if other than specified (see 6.7.2).

k. That the safety precaution requirements of DOD 4145.26-M, DoD Contractor's Safety Manual for Ammunition and Explosives, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of OP 5, Ammunition and Explosives Ashore, are applicable.

6.3 First article. When a first article sample is required, the contracting officer should provide guidance as to the number of items to be tested as specified in 4.3 and whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has previously been acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids except when specified (see 6.2).

6.4 Propellant history. The nominal values shown in table VII are based on formulations used by Naval Surface Warfare Center Indian Head Division (NSWC IHD). NSWC IHD, which uses Baker-Perkins brand vertical mixers, has successfully met the requirements of this specification using those values. Due to the known effects of moisture-related degradation of AP-based composite propellants, manufacturing processes should be carefully designed to severely restrict the propellant and ingredients exposure to moisture and contaminants prior to unit loading (sealing off the propellant).

TABLE VII. Properties of NSWC/IH-BC-16 propellant

Theoretical Property	
FLAME TEMPERATURE (°F, CALCULATED), CHAMBER GLASS TRANSITION TEMPERATURE (°F) SPECIFIC HEAT RATIO (GAMMA, CALCULATED)	4215 -97 1.21
SPECIFIC IMPULSE (1000 PSIA IN STANDARD ATMOSPHERE, LBF-S/LBM, CALCULATED	235.9
TEMPERATURE SENSITIVITY (%/°F) RANGE OF -65 TO 165 °F	0.043

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TABLE VII. Properties of NSWC/IH-BC-16 propellant – Continued.

Approximate Exhaust Gas Composition (Mole %, 1Atm.):					
H_2O	27.13	N_2	8.01	H	0.20
CO	24.04	CO_2	5.36	C1	0.11
H_2	19.05	$A1_2O_3$ (s)	0.70	SiO (s)	0.08
HC1	14.73	$FeC1_2$	0.43	HO	0.06
Molecular weight of exhaust				23.59	

6.5 Burn rates. Past acceptable lots of propellant have conformed to the burn rates of tables III, IV, V, and VI.

6.6 Propellant properties. The properties of the NSWC/IH-BC-16 propellant are shown in table VII.

6.7 Definitions.

6.7.1 Batch. A batch is a quantity of propellant consisting of a mix of all the ingredients including curative manufactured at one time in one mixer per the contractor's normal practice.

6.7.2 Lot. Unless otherwise specified (see 6.2), each batch of propellant should constitute a lot for inspection purposes. No batch of propellant should contain more than one batch (or lot) of any ingredient.

6.7.3 Process change. A process change is a change in the way the propellant is manufactured, the way its ingredients are manufactured, or both. Process changes include, but are not limited to, a change in:

a. Ingredient manufacturer, or the manufacturer's process

b. Mixing vessel size or type

c. Mix time or temperature

d. Cure temperature

6.8 Packaging. When packaging is required for propellant shipment, the propellant should be individually wrapped in moisture proof packing material and properly identified as to propellant lot and batch number.

6.9 Curing hardness test. Using a small plastic beaker of propellant as a curing sample, the following tests will be performed to determine when the propellant has attained sufficient cure for further processing. A cut section of a pan or carton sample may also be used.

a. Use a craftsman knife, exacto knife, or equivalent to obtain a cut surface layer of the propellant over an area sufficient to squarely support the Shore "A" gage presser foot. Do not use the same place on the sample for readings done on subsequent days.

b. Perform the Shore "A" durometer test at 140±5 °F in accordance with ASTM D2240, method A. Reading is to be taken 10±1 seconds after presser foot is in contact with specimen.

c. When two measurements taken 24 hours apart show less than a 2 durometer point increase with time, then the sample should be deemed sufficiently cured.

d. When the two measurements 24 hours apart indicate a decrease in hardness, an additional wait of 24 hours is required to ensure that cure has progressed to the point that another reading may be taken.

6.10 Subject term (key word) listing.

AEPS
Aircraft Ejection Seat Catapults
Cartridges
CCU-22B/A
CCU-149/A
CKU-5C/A
CKU-7A/A
CKU-10A/A

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CKU-11A/A

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2025-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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